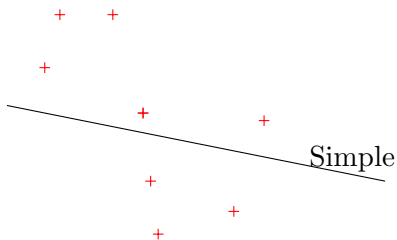


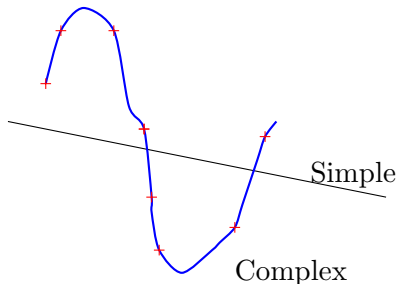
Module : Bias and Variance



The points were drawn from a sinusoidal function (the true $f(x)$)

- Let us consider the problem of fitting a curve through a given set of points
- We consider two models :

$$\begin{matrix} \textit{Simple} \\ \textit{(degree:1)} \end{matrix} \quad y = \hat{f}(x) = w_1x + w_0$$



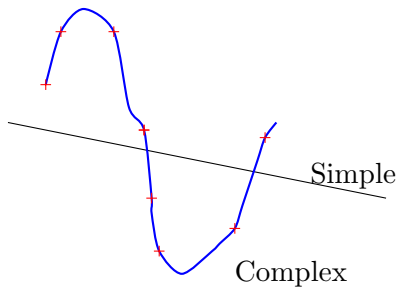
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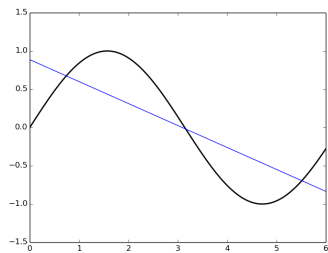
$$\begin{matrix} \text{Complex} \\ (\text{degree:25}) \end{matrix} \quad y = \hat{f}(x) = \sum_{i=1}^{25} w_i x^i + w_0$$

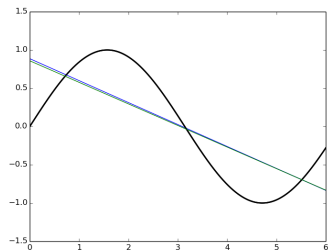
- Note that in both cases we are making an assumption about how y is related to x . We have no idea about the true relation $f(x)$
- The training data consists of 100 points

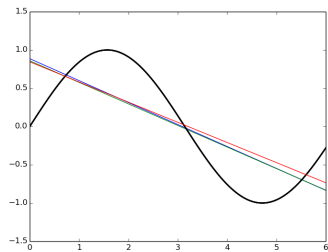


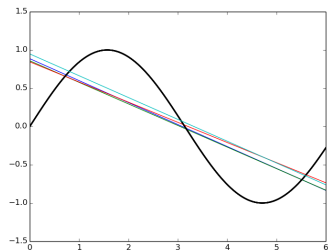
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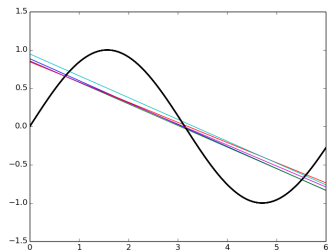
- We sample 25 points from the training data and train a simple and a complex model
- We repeat the process ' k ' times to train multiple models (each model sees a different sample of the training data)
- We make a few observations from these plots

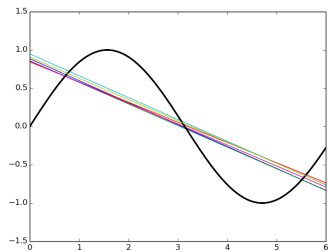


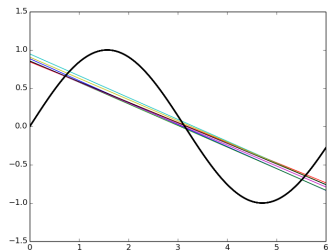


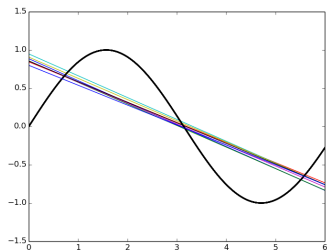


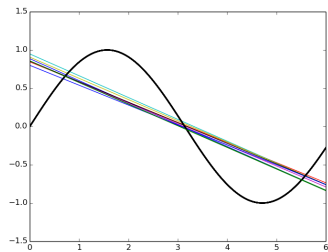


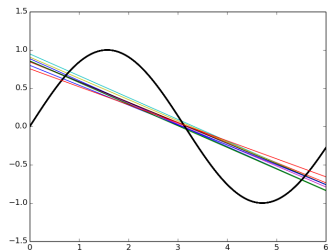


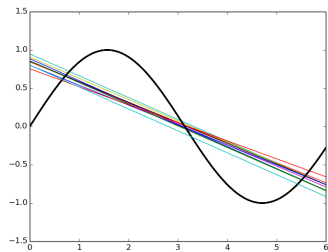


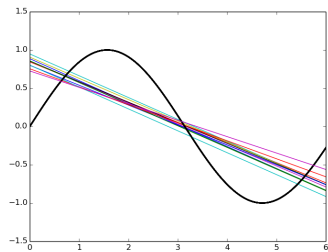


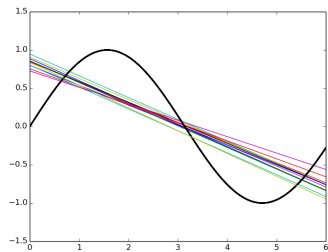


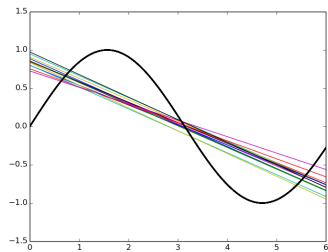


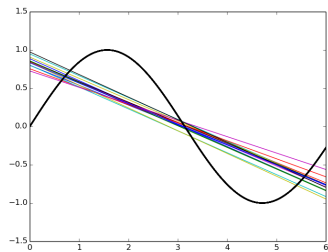


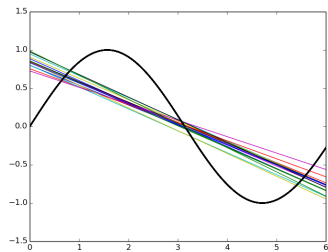


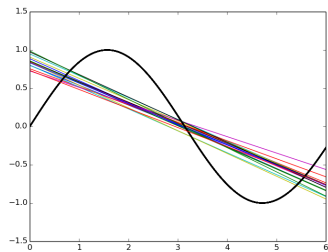


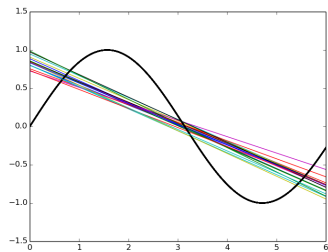


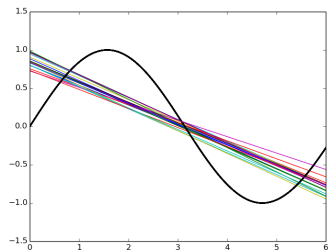


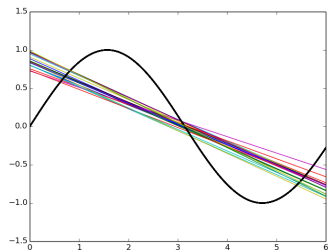


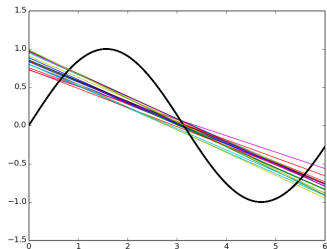




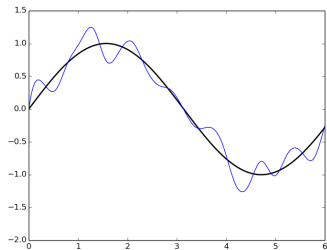
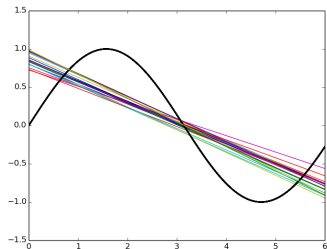




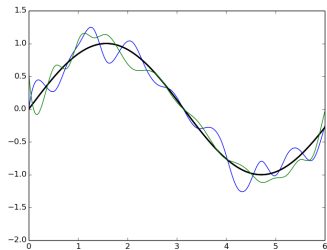
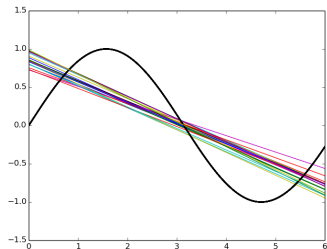




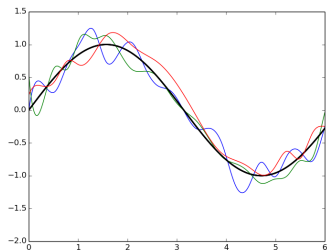
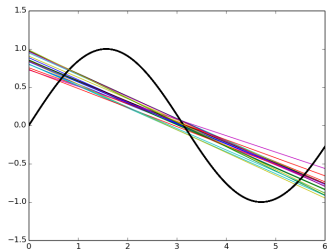
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- However they are very far from the true sinusoidal curve (under fitting)



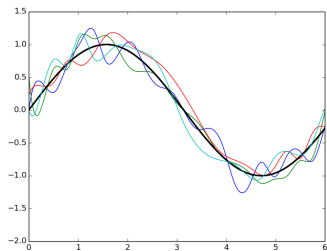
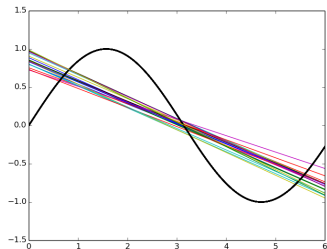
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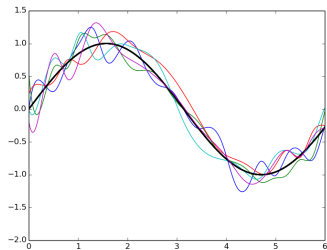
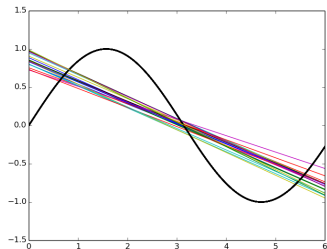
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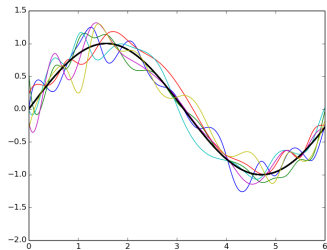
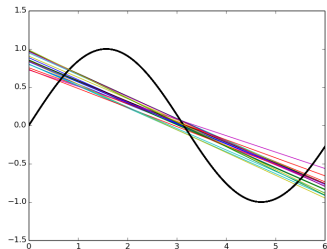
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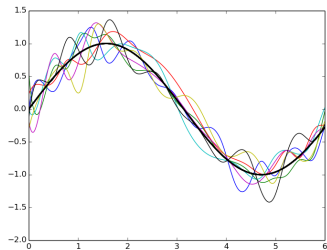
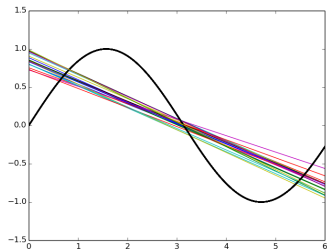
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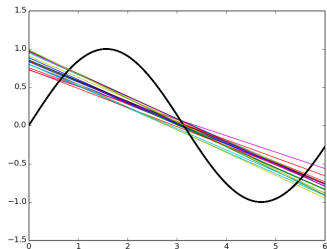
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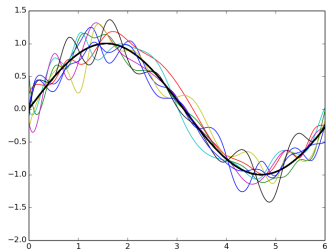
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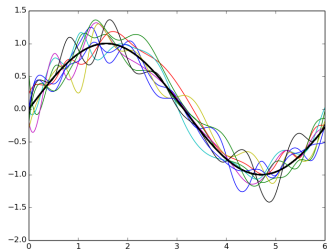
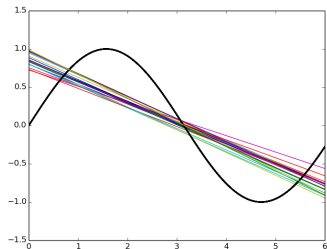


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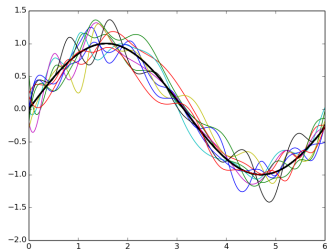
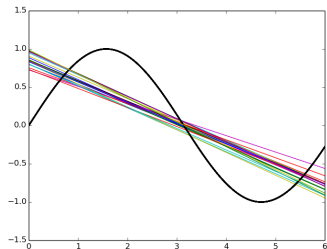


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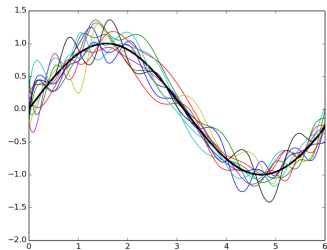
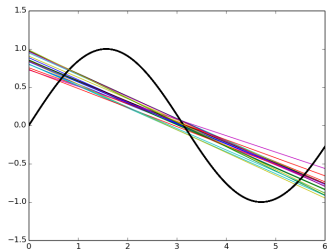




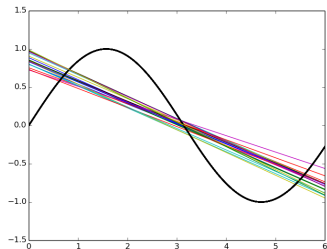
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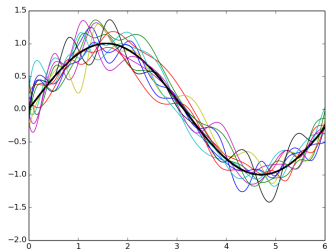
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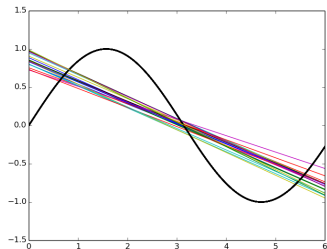


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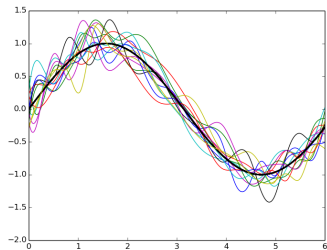


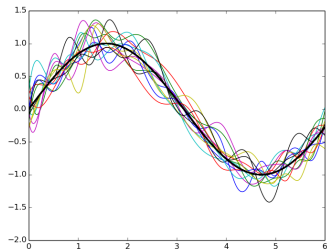
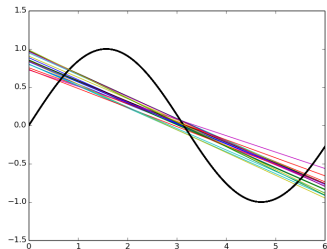
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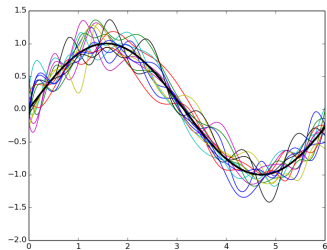
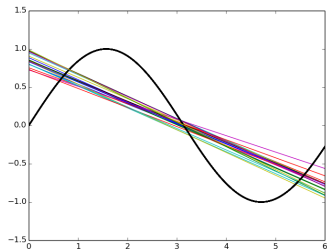


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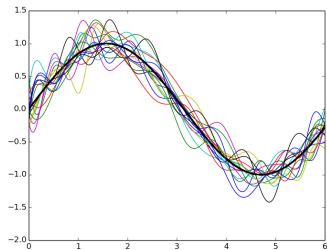
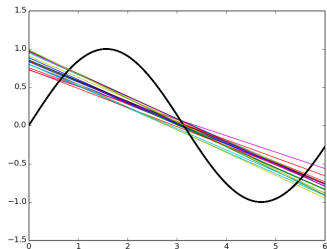




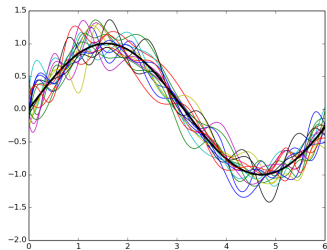
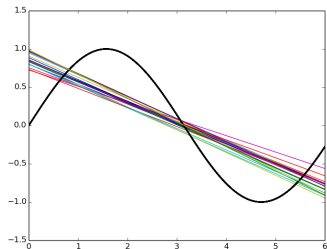
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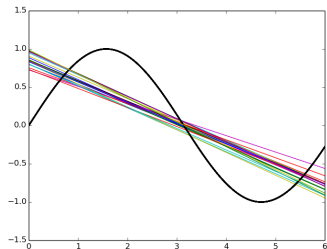
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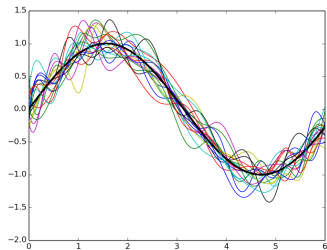
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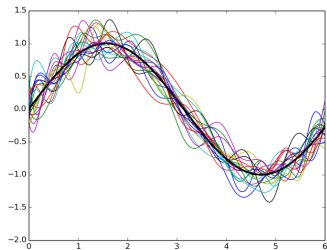
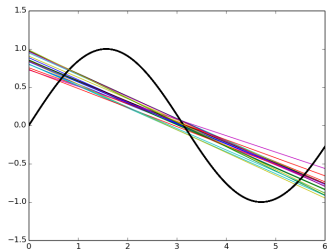


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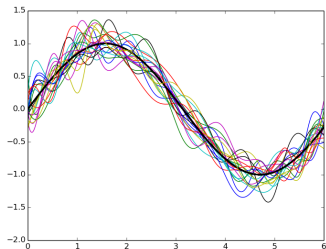
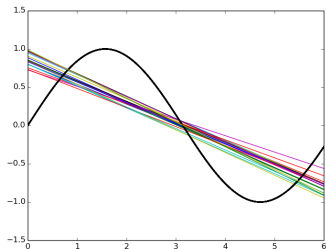


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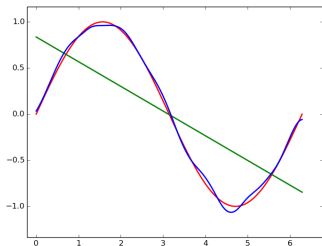




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- Simple models trained on different samples of the data do not differ much from each other
- However they are very far from the true sinusoidal curve (under fitting)
- On the other hand, complex models trained on different samples of the data are very different from each other (high variance)



Green Line: Average value of $\hat{f}(x)$ for the simple model

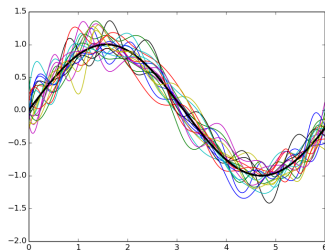
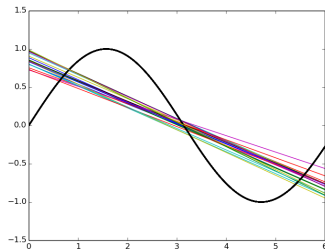
Blue Curve: Average value of $\hat{f}(x)$ for the complex model

Red Curve: True model ($f(x)$)

- Let $f(x)$ be the true model (sinusoidal in this case) and $\hat{f}(x)$ be our estimate of the model (simple or complex, in this case) then,

$$\text{Bias}(\hat{f}(x)) = E[\hat{f}(x)] - f(x)$$

- $E[\hat{f}(x)]$ is the average (or expected) value of the model
- We can see that for the simple model the average value (green line) is very far from the true value $f(x)$ (sinusoidal function)
- Mathematically, this means that the simple model has a high bias
- On the other hand, the complex model has a low bias

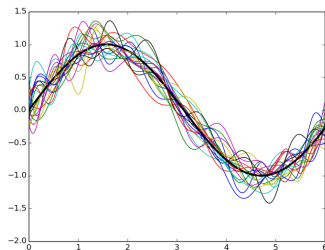
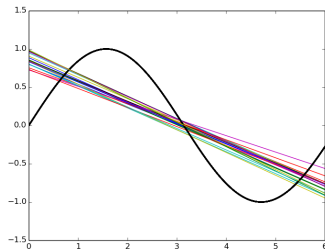


- We now define,

$$\text{Variance } (\hat{f}(x)) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

(Standard definition from statistics)

- Roughly speaking it tells us how much the different $\hat{f}(x)$'s (trained on different samples of the data) differ from each other
- It is clear that the simple model has a low variance whereas the complex model has a high variance



- In summary (informally)
- Simple model: high bias, low variance
- Complex model: low bias, high variance
- There is always a trade-off between the bias and variance
- Both bias and variance contribute to the mean square error. Let us see how

- **Bias** measures how far our estimated model is from true model.
- **Variance** measures how much our estimated model $\hat{f}(x)$ varies across different training datasets.
- Our estimate $\hat{f}(x)$ means the function that we learn from data

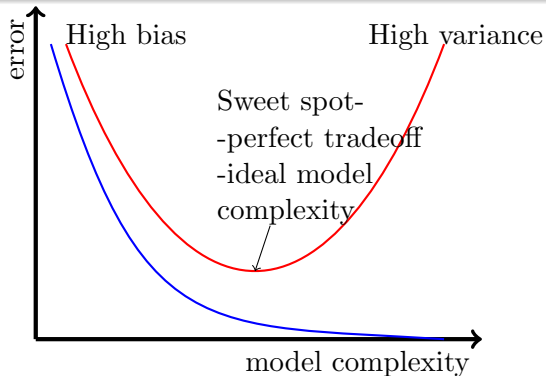
- We can show that

$$\begin{aligned} E[(y - \hat{f}(x))^2] &= \textit{Bias}^2 \\ &+ \textit{Variance} \\ &+ \sigma^2 \text{ (irreducible error)} \end{aligned}$$

- Consider a new point (x, y) which was not seen during training
- If we use the model $\hat{f}(x)$ to predict the value of y then the mean square error is given by

$$E[(y - \hat{f}(x))^2]$$

(average square error in predicting y for many such unseen points)



$$E[(y - \hat{f}(x))^2] = Bias^2 + Variance + \sigma^2 \text{ (irreducible error)}$$

- The parameters of $\hat{f}(x)$ (all w_i 's) are trained using a training set $\{(x_i, y_i)\}_{i=1}^n$
- However, at test time we are interested in evaluating the model on a validation (unseen) set which was not used for training
- This gives rise to the following two entities of interest:
train_{err} (say, mean square error)
test_{err} (say, mean square error)
- Typically these errors exhibit the trend shown in the adjacent figure

Intuitions developed so far

- Let there be n training points and m test (validation) points

$$train_{err} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$test_{err} = \frac{1}{m} \sum_{i=n+1}^{n+m} (y_i - \hat{f}(x_i))^2$$

- As the model complexity increases $train_{err}$ becomes overly optimistic and gives us a wrong picture of how close $\hat{f}$ is to f
- The validation error gives the real picture of how close $\hat{f}$ is to f